

Multi-Source Data Scraping & Trust Scoring System

1. Scraping Strategy

The system collects structured data from three sources: blogs, YouTube, and PubMed. Each source is handled using an approach suited to its format.

For blogs, content is extracted using BeautifulSoup by parsing HTML elements and metadata. Irrelevant elements such as navigation bars, advertisements, and UI noise are removed to ensure clean text extraction.

For YouTube, metadata (title, channel, publish date) is retrieved using yt-dlp. Transcripts are fetched using the YouTube Transcript API. If unavailable, video descriptions are used as fallback. Additional preprocessing removes timestamps, URLs, and noise.

For PubMed, the Entrez API (BioPython) is used to reliably extract structured scientific data such as authors, journal, abstract, and publication year.

All data is standardized into a unified JSON schema to ensure consistency across sources.

2. Topic Tagging Method

Topic tagging is implemented using a rule-based keyword matching approach. A predefined set of keywords (e.g., AI, machine learning, NLP, healthcare) is used to identify relevant topics in the content.

The text is normalized and scanned for keyword matches, which are added to the topic_tags field. If no keywords are detected, a fallback tag ("general") is assigned.

This approach is efficient, interpretable, and works well across different content types without requiring additional models.

3. Trust Score Algorithm

The trust score evaluates content reliability on a scale from 0 to 1 using a weighted combination of factors:

Trust Score =

$0.25 \times \text{Author Credibility} +$

$0.20 \times \text{Citation Score} +$

$0.20 \times \text{Domain Authority} +$

$0.20 \times \text{Recency} +$

$0.15 \times \text{Medical Disclaimer Presence}$

- **Author Credibility:** Higher for multiple or known authors; lower for missing authors.
- **Citation Score:** Highest for PubMed (peer-reviewed), moderate for YouTube, lower for blogs.

- **Domain Authority:** Trusted domains receive higher scores; unknown domains are penalized.
- **Recency:** Recent content is prioritized over older content.
- **Medical Disclaimer:** Content with safety disclaimers is considered more reliable.

The final score is normalized to remain within [0, 1].

4. Edge Case Handling

The system handles several edge cases to ensure robustness:

- Missing metadata → default values (“unknown”)
- Missing transcripts → fallback to description or title
- Multiple authors → higher credibility scoring
- Non-English content → automatic language detection
- Empty content → fallback mechanisms applied
- Long content → split into smaller chunks

These mechanisms ensure stable and consistent outputs across all sources.

Conclusion

This project demonstrates a practical data pipeline that integrates multiple sources, processes unstructured data, and evaluates content reliability using a structured trust scoring system. The design focuses on robustness, modularity, and real-world applicability.